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Factors influencing home advantage in American collegiate football

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ABSTRACT

This study investigated the effect of home advantage (HA) in NCAA Division I football, and assessed factors related to why HA may exist. Game location (home/away), points differential between team scores, distances between team stadiums (short [<483 km], moderate [484–836 km], long [837–1331 km] or very long [>1332 km]), crowd number, differences in team penalty counts (penalty differential), penalty yardage differential and time zone change (Eastward or Westward direction of travel) data from two non-consecutive NCAA college football seasons (2013 and 2016; 719 games; 222 teams) were obtained. Matches played at home were associated with a ~5 point advantage (ES = 0.58 ± 0.18). Away teams traveling >484 km resulted in a moderate disadvantage (~5 to 7 points, ES = −0.61 to −0.77). When crossing time zones in a Westerly direction resulted in a ~7.5 point disadvantage (ES = −0.82 ± 0.31) for away teams. When playing at Home, larger crowds resulted in a ~6 point advantage (ES = 0.67 ± 0.36), whilst larger crowds were negatively associated with performance (~8 points; 2 points) for away teams (ES = −0.97 ± 0.27). These results suggest HA exists in NCAA Division I football, particularly evident with larger crowds. These results provide evidence that could aid logistical or periodisation strategies in preparation for games.

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Gridiron; recovery; athlete; performance; travel; United States

Introduction

Home advantage (HA) represents the observation that the number of points scored/games won in sports competitions played at home is greater than those played away (Nevill and Holder 1999; Forrest et al. 2005; Pollard and Pollard 2005). The presence of HA in professional sport is a consistent and robust finding within many professional sports (Pollard 1986; Courneya and Carron 1992; Jones 2017). For example, HA has been reported in soccer (Clarke and Norman 1995), basketball (Snyder and Purdy 1985), baseball (Winter et al. 2009), rugby (Thomas et al. 2008), tennis and golf (Nevill et al. 1995); though the extent of its existence varies based on the specifics of the particular sport. For instance, baseball, basketball, and hockey show a greater effect for playing at home than American football at professional levels (Pollard and Gómez 2015), potentially due to a larger crowd presence and/or increased travel demands of these competitions. Whilst HA is clearly evident in professional sports, few peer-reviewed resources assess the existence of HA in American collegiate football (ACF). This is surprising given ACF is one of the most popular and wealthiest sports in the world, and played in a large and diverse country. Further research which determines the existence of HA in ACF, along with potential factors underpinning such an existence, would appear warranted.

The "home effect" in ACF has been hypothesised to exist due to a number of interrelated factors including travel distance, strength of schedule/opponent, injury status of the team, time zone change and game time, referee bias and crowd size (Nevill and Holder 1999). Indeed, given the difference in resources and vast number of teams within National Collegiate Athletic Association (NCAA) football (128 teams), large discrepancies exist in the quality of opponents along with crowd size variation from 10,000 to 120,000 people (Fullagar et al. 2017). Crowd noise has been shown to have an influence on referring decisions (i.e., home teams receive more favourable decisions) which can potentially impact match outcome (Carron and Paradis 2014). Further issues with determining the extent of HA within ACF may exist due to scheduling and travel schedules creating large variability in team playing schedules as a result of the NCAA rotational policy (Redding 2017). For example, due to the geographical location of in- and out-of-conference teams, some teams will travel short compared to long distances prior to games. Proposed mechanisms behind the disadvantages of travel on match performance include athletes encountering greater perceived fatigue and poorer mood states, resulting in greater perceived vulnerability during away competitions (Carron and Paradis 2014). With the many interplaying factors that may affect HA in ACF, quantification of the presence and effectors would appear important to benefit both ACF researchers and practitioners alike.

Of the limited literature that exists, a 6-point advantage has been reported for home teams compared to away teams during 661 Division I collegiate football matches played over two seasons (Wang et al. 2011). In addition, these authors found that higher strength opponents (Bowl Championship Series (BCS) teams)) exhibit a 7-point advantage over weaker opponents (non-BCS teams). These findings of an HA existed when controlling for game location, level of opponent, penalty differential, attendance and travel distance. For example, an increase in crowd size has been shown to improve a side's chances of

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winning for home teams (Schwartz and Barsky 1977), whilst the crowd itself has been hypothesised to bias referee decisions in favour of the home team (Nevill and Holder 1999; Downward and Jones 2007). However, there were no analyses on direction of travel in this study. This is surprising given the potential differences between travelling east vs. west on game outcomes (Brager and Mistovich 2017), along with greater detrimental effects on maximal and intermittent-sprint performance following (prolonged) eastward travel (Fowler et al. 2017). Indeed, this may have flow on effects on team's scoring opportunities and ultimately overall game outcome (Carron and Paradis 2014). In addition, this work investigated games from nearly a decade ago. Since this time, there have been dramatic changes within collegiate football including increases in television audiences and engagement, financial resources and recruiting (Fullagar et al. 2017). Although speculative, it may be possible that, like evidence in other sports (ice hockey, and basketball; (Pollard and Pollard 2005)), HA may have declined over the past decade.

Taken collectively, more evidence is required to confirm the assumption that an HA remains present in collegiate football and what antecedents exist to promote the presence of HA in ACF, i.e., travel, crowd size, or officiating bias. Accordingly, the aim of this study was to determine the effect of HA in NCAA Division I football, and assess factors related to why HA may exist.

Methods

Data set

Data from two non-consecutive college football seasons (to avoid any bias based on roster strength) were drawn from a commercial statistics provider (2013 and 2016; Sport Source Analytics) by the authors. This included match description and outcome data for every team that played a game within the NCAA Division I competition. Specifically this included passing and rushing yards gained, penalty counts and penalty yards conceded, along with crowd size location and if the game was in- or out-of-conference. A subset of in-conference games played at non-neutral venues was used, which included 719 games across the two seasons for 222 teams. Additional information was added using Google Maps to calculate the travelling distances between the stadiums of the teams (i.e., the home location was set as the Universities home stadium). The *geonames* package in R was used to identify the time zone information based on the home city. Since data are retrospective, publicly available and is from a team perspective (rather than release of individual data), no ethics was required for this investigation.

Study design

Within NCAA competition, end-of-season rankings are determined subjectively by a panel of experts (i.e., the College Football Playoff Committee (Playoff 2012)), and post-season matches are allocated as a result, with the four highest ranked teams competing for the National Championship (Redding 2017). Typically, the teams with the most amount of wins are ranked the highest but in the event in of teams achieving similar win/loss records, the panel may decide on final rankings not only from wins/loss records but also the magnitude of these results, strength of opponents and whether they won their division/conference championship (Playoff 2012). Given such premise, this study considered various quantifiable aspects of college football that are hypothesised to contribute to HA. Primarily, this included points' differential at the completion of each match as the primary outcome measure. Additionally, differentials in both penalty counts (penalty differential) and penalty yardage (penalty yard differential) were calculated as secondary outcome measures using the totals obtained in the main data set. Matches were grouped by location as either Home or Away. Away matches were further classified depending on the distance and direction of travel. To quantify the contribution of the effect of travel on HA, travel distances were grouped by quartiles as Short (<483 km), Moderate (484–836 km), Long (837–1331 km) or Very Long (>1332 km), or number of time zones covered in either an easterly or westerly direction. Lastly, crowd number for each match (from the obtained data set) was used to assess the associations between differing crowd sizes and HA.

Statistical analyses

The effect of an HA was assessed using linear mixed models. In these models, individual teams were specified as a random effect to account for different within-team standard deviations (SD), and fixed effects included each of the selected match statistics. Prior to the construction of each model, within- and between-subject SD were determined for each outcome/match statistic using a mixed-model reliability analysis, using the variance of the random effect and the residual, respectively. For factor predictor variables (i.e., match location) the slope of the model represented the observed effect in the presence of that variable (i.e., Home). For models incorporating a continuous predictor, the slope of the model was multiplied by two SD of the predictor, representing a change from a typically low to a typically high value. All outcomes were converted to an effect size (ES) and the associated 90% confidence limits (90% CL) by dividing by one between-team SD of the outcome variable (i.e., points differential, penalty differential or penalty yard differential). ESs were interpreted as <0.2, trivial; 0.2–0.6, small; 0.6–1.2, moderate; 1.2–2.0, large; and >2.0, very large (Hopkins 2004; Batterham and Hopkins 2006). Effects were considered real if the likelihood of the effect exceeded 75% (Hopkins 2004; Batterham and Hopkins 2006). All statistical analyses were performed using RStudio (v. 3.4.1). Results are presented as mean; 90% CL, unless otherwise stated.

Results

Home advantage and points differential

The effect of HA is shown in Table 1. Matches played at home were associated with a ~5.3 point advantage (ES = 0.58 ± 0.18). No clear differences between Home and Away teams were observed for either penalty differential or penalty yard differential.

Table 1. The effect of home advantage, travel and crowd in American collegiate football.

Model	Difference	CI (±)	ES	CI (±)	Likelihood (%)
Home ground advantage					
Points diff & location (home)	5.34	1.66	0.58	0.18	100
Penalty yards diff & location (home)	−1.60	2.85	−0.38	0.67	67
Penalty count diff & location (home)	−0.17	0.31	−0.37	0.68	66
Effect of travel					
Points diff & travel					
Short (<483 km)	−2.10	2.75	−0.23	0.30	56
Moderate (484–836 km)	−6.31	2.66	−0.68	0.29	100
Long (837–1331 km)	−5.68	2.67	−0.61	0.29	99
Very long (>1332 km)	−7.18	2.72	−0.77	0.29	100
Points diff & crossing a TZ	−5.15	2.11	−0.56	0.23	99
Points diff & crossing a TZ East	−2.69	2.91	−0.29	0.31	68
Points diff & crossing a TZ West	−7.57	2.88	−0.82	0.31	100
Effect of crowd					
Points diff & crowd (away team)	−8.69	2.40	−0.97	0.27	100
Points diff & crowd (home team)	6.12	3.29	0.67	0.36	98
Points diff & 10,000 fans (away team)	−4.12	1.14	−0.46	0.13	100
Points diff & 10,000 fans (home team)	2.36	1.27	0.26	0.14	76

Abbreviations: CI; confidence interval, diff; differential, ES; effect size, km; kilometres, TZ; time zone.

Home advantage and travel

Away teams traveling <483 km were not substantially affected (Table 1; Figure 1), while all travel distances >484 km resulted in a moderate disadvantage (~5 to 7 points, ES = −0.61 to −0.77 ± ~0.29). Away teams that were required to cross at least one time zone experienced a ~5 point disadvantage (ES = 0.56 ± 0.23). There was no clear disadvantage experienced by teams traveling East (ES = −0.29 ± 0.31), but crossing time zones in a Westerly direction resulted in a ~7.5 point disadvantage (ES = −0.82 ± 0.31).

Home advantage and crowd size

When playing at Home, larger crowds resulted in a ~6 point advantage (ES = 0.67 ± 0.36), whilst bigger crowds were negatively associated with performance (−8.69 points; 2.40 points) for teams away from home (ES = −0.97 ± 0.27). When adjusted to the effect per 10,000 people, Home teams were advantaged by ~2 points (ES = 0.26 ± 0.14), and away teams were disadvantaged by ~4 points (ES = −0.46 ± 0.13).

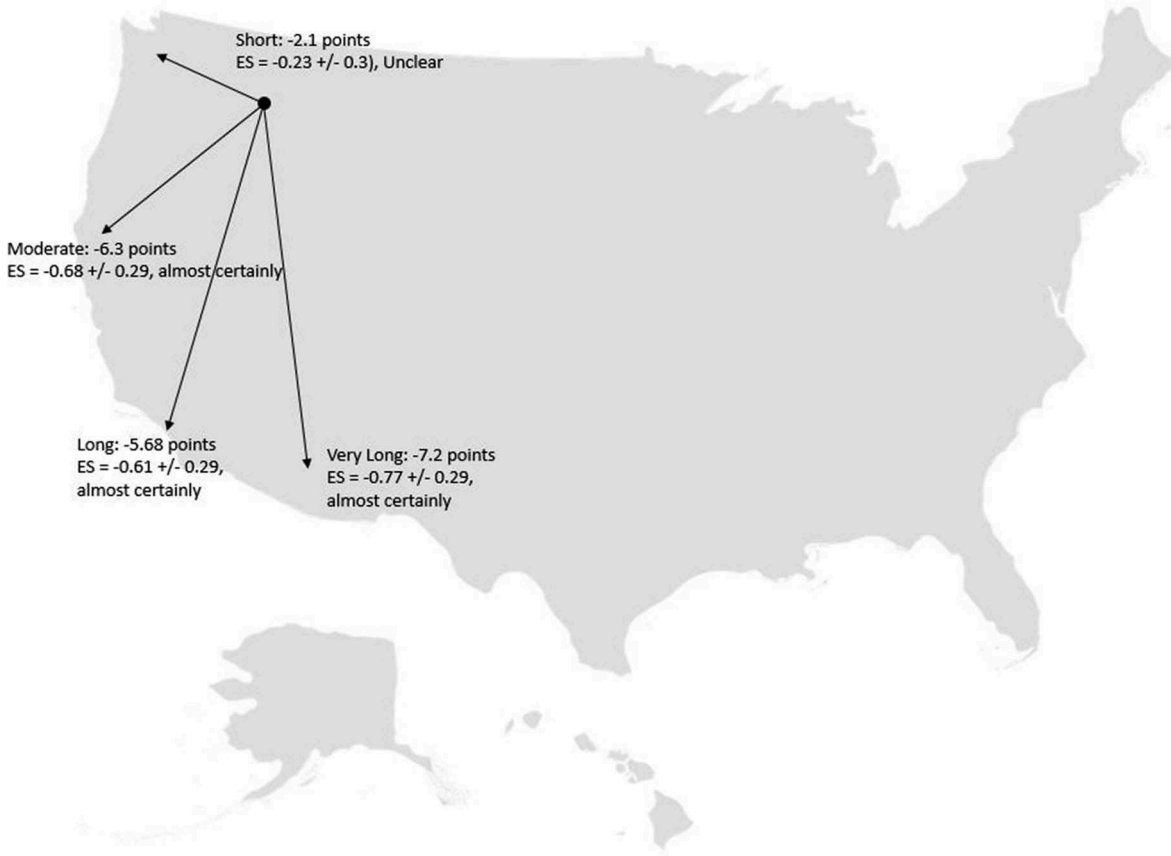

Figure 1. Visual representation of home advantage in American collegiate football.

Discussion

This study investigated the effect of HA in NCAA Division I ACF football, and assessed factors related to why HA may exist. The main finding was that playing at home is worth an additional 5 points. Whilst there was no disadvantage when playing away from home when travel was relatively short (<483 km), further travel resulted in a ~-6-point differential; which was further exacerbated when traveling in a westerly direction with changes in time zones (~7.5 points). Additionally, a 2 SD change in crowd was worth ~11 points to the opposition when playing away, and worth ~6 points when playing at home. We recommend these findings be treated with caution due to the small ES; however, we note these results would be meaningful to the American Football coaching population due to scoring context of ~7 points (i.e .a touchdown). Taken collectively, these results provide further evidence of the extent of HA in ACF and importance of crowd and travel to match outcomes; both of which could aid logistical or periodisation strategies in preparation for games (Robertson and Joyce 2015).

Previous research has shown that an HA exists in sports including soccer (Clarke and Norman 1995), basketball (Snyder and Purdy 1985), baseball (Winter et al. 2009), rugby (Thomas et al. 2008), tennis and golf (Nevill et al. 1995); however, there is currently limited evidence of such a relationship in ACF (Wang et al. 2011). The main finding from the current study was that playing at home is worth an additional 5 points. This may be considered a fairly small margin compared to a scoring outcome (i.e., less than a touchdown plus extra point [7 points]). However, interestingly the present HA is almost double the average home team point differential for the seasons analysed (2.7 and 2.9 points for 2013 and 2016, respectively; field goal is worth 3 points) providing context for the importance of HA. Similarly, in an examination of HA from 661 ACF regular season games during the 2008–2009 NCAA Division IA season, it has been shown that the advantage of playing at home was ~6 points and the disadvantage of playing away was ~3 points (Wang et al. 2011) as based on travel distance as a predictor. These authors hypothesised that intangible causes such as environmental familiarity and psychological factors related to game situations (pressure, rewards) may be responsible for the prevalence of an HA. For instance, in an archival analysis of baseball data from World Series from 1924 to 1982, Baumeister and Steinhilber (Baumeister and Steinhilber 1984) found that when home teams played in a decisive Championship Game Seven they won only 39% of the games. However, such a concept is less likely in our study since only in-season games were analysed (at non-neutral venues). Alternatively, although difficult to quantify in our study, it is possible that an improved psychological and behavioural state, whereby players may have had enhanced perceptions of confidence and motivation when playing at home (Tenenbaum 1996), contributed to the 5-point advantage at home. Nonetheless, this remains speculative and further research that quantifies such psychological associations and comparisons to HA outcomes is required.

The present study provides additional context to the presence of HA in ACF, as while there was no disadvantage when playing away from home involving relatively short travel (<483 km), a ~6 point disadvantage was evident for travel >483 km. Previous research investigating the effect of travel distance on HA is conflicting. A study of 3496 football games (soccer; English Football league, European Cup, FA Cup) reported no differences in HA between teams travelling over or under 200 miles (Pollard 1986). In contrast, a 25% improvement in home winning percentage was reported in collegiate basketball teams when the away teams travel more than 200 miles (Snyder and Purdy 1985). Appreciating the sport-specific context of travel and scheduling that may cloud this issue, with regard to American football, our findings conflict with those from the National Football League (NFL) (Lopez et al. 2017). Specifically, no significant HA exists in the NFL (Lopez et al. 2017), but evidence from collegiate football shows the disadvantages of travelling, in that the HA is most pronounced when the visiting team have the longest travel distance (Wang et al. 2011). Therefore, it would appear that travel distance may be a significant predictor of HA in collegiate football. Indeed, there may be differing effects across conferences within ACF (e.g., average travel distance, kick off time), which may merit further investigation.

There is an assumption that the predominant mechanism behind the disadvantage of travelling is due to fatigue; however, the limited evidence suggests that short-haul travel in team sport (air travel for ~5–6 h) has negligible effects on player preparedness (Fowler et al. 2015a), wellness (Fowler et al. 2015b) or sleep quantity and quality, hydration, and perceptual fatigue (Fowler et al. 2014). It is possible that when travel was extended beyond 483 km in our study that familiar routines or habits of players were disrupted, causing a discomfort response which may have impacted performance (Tenenbaum 1996). Indeed, the age cohort (18–22 years) in our study may be still developing routines compared to older, more experienced professional players (i.e., NFL players). This may leave them more susceptible to negative influences from travel or changes in their preparation. For example, it has previously been shown that subjective jet-lag after travelling over one time zone was misinterpreted as increased fatigue due to sleep disruption in less-experienced players (Fowler et al. 2015a). This perception may have impacted upon player preparedness especially in less experienced players in our study.

In our investigation, there was no clear disadvantage imposed on teams crossing time zones in an easterly direction, though traveling west resulted in a ~7.5 point disadvantage (ES = −0.82 ± 0.31). Previous authors have examined the effect on HA when travelling across time zones. Jehue et al. (1993) investigated the effect of time zone and game time changes on 27 NFL teams' performance over a 10-year period. West coast teams travelling to central or eastern venues were shown to incur performance decrements when playing during the day, whilst when playing at home or away at night, West coast teams showed a performance improvement. This supports the small effect of away teams incurring a disadvantage when travelling west for competition. Fowler and colleagues (Fowler et al. 2017) reported that long-haul transmeridian eastward travel can impede optimal sleep, subjective jet lag, fatigue, and motivation as well as the recovery of maximal and intermittent sprint performance in physically trained males. Although long-haul travel (>6 h) is not present in the current study, our results are in opposition to previous research which

shows the advantage of westbound travel compared with eastbound travel (Jehue et al. 1993; Brager and Mistovich 2017; Fowler et al. 2017). The difference in our results is likely due to the majority of games (~95%) studied endure only one time zone shift, limiting the inference of a small sample games which show a disadvantage of travelling west. It is also possible that the strength of west teams was greater than east teams, and/or crowd support is stronger in the west in the time period studied.

A 2 SD change in crowd (~60 000 people) was found in our study to be worth 11 points to the opposition when playing away (ES = 0.51), and worth 6 points (ES = 0.29) when playing at home. The advantages of crowd size on HA have been hypothesised to be due to numerous factors such as positive support (i.e., cheering) for the home team, negative support for the away team (i.e., booing), influence on referee decision-making (i.e., greater or lesser penalties) and intimidation of the away team (i.e., distraction/crowd noise) (Nevill and Holder 1999). However, due to larger crowds being common to higher ranked teams in the sport studied in our investigation, in such extreme examples it is likely success may come before the crowd, not the other way around. Alternatively, one can consider the effect on points differential in smaller crowd increments (i.e., 10 000 fans). For instance, in our study when away teams had an extra 10 000 fans they incurred a 2-point disadvantage, whilst to the home team 10 000 fans was worth an extra 1.2 points. Thus, when combined, having an additional 10 000 people could represent the advantage of a field goal (3 points), which although seemingly small would presumably be of interest for American Football coaches. Indeed, there are example in other sports where crowd saturation or density matters to a point before being redundant. For example, in a study in baseball HA went from 48% then stadia were less than a fifth full to 57% when over 40% full (Schwartz and Barsky 1977). However, there is also opposing evidence with marginal differences in HA in the top four English football (soccer) divisions, even though crowd density ranged from 20–70% across divisions (Dowie 1982; Pollard 1986). Thus, it would appear that the effect of crowd on HA is context and sport dependant. For instance, it has been shown that the variance in crowd behaviour can positively and negatively influence HA (normal crowd behaviour benefits the home team, whilst antisocial crowd behaviour, such as swearing, benefits the away team (Thirer and Rampey 1979)).

Conclusion

Playing at home in NCAA Division I football is worth an additional 5 points. There was no disadvantage when playing away from home when travel was relatively short (<483 km), however it was worth ~6 points for anything >483 km. In addition, westerly travel appears more disadvantageous than easterly travel. Collectively these results further our understanding of the effects of HA in collegiate football and potentially aid preparation strategies practitioners can undertake for their athletes.

Perspective

Our findings support an existence of HA in ACF, which displays the difficulties for teams when they are forced to travel >483 km. Further research which depicts whether the primary affect is an advantage for the home team or a disadvantage for the away team would help identify further underlying causes which contribute to this home and crowd advantage, and travel disadvantage. Teams whom appear to be disadvantaged may combat these situations by using this information to alter their preparation strategies and coaching/playing style (e.g., using techniques to combat crowd noise throughout the preceding training week). However, one should consider the numerous controllable and uncontrollable factors that could affect this (i.e., days between games, training programmes, school schedule, level of experience of players, overall health of team, time of season).

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Competing interest

The authors do not have professional relationships with any companies or manufacturers that will benefit from this study. The authors declare no conflicts of interest.

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References

Batterham A, Hopkins W. 2006. Making meaningful inferences about magnitudes. Int J Sports Physiol Perform. 1:50–57.
Baumeister R, Steinhilber A. 1984. Paradoxical effects of supportive audiences on performance under pressure: the home field disadvantages in sports championships. J Personal Soc Psychol. 47:85–93.
Brager A, Mistovich R. 2017. Game times and higher winning percentages of west coast teams of the National Football League correspond with reduced prevalence of regular season injury. J Strength Cond Res. 31 (2):462–467.
Carron A, Paradis K. 2014. The home advantage. In: Eklund, R, Tenenbaum, G, editors. Encyclopedia of sport and exercise psychology. Thousand Oaks (CA): Sage Publications; p. 301–355.
Clarke S, Norman J. 1995. Home ground advantage of individual clubs in English soccer. Statistician. 44:509–521.
Courneya K, Carron A. 1992. The home advantage in sport competitions: a literature review. J Sport Exerc Psychol. 14:13–27.
Dowie J. 1982. Why Spain should win the World Cup. New Scientist. 94:693–695.
Downward P, Jones M. 2007. Effects of crowd size on referee decisions: analysis of the FA Cup. J Sports Sci. 25(14):1541–1545.
Forrest D, Beaumont J, Goddard J, Simmons R. 2005. Home advantage and the debate about competitive balance in professional sports leagues. J Sports Sci. 23(4):439–445.
Fowler P, Duffield R, Howle K, Waterson A, Vaile J. 2015a. Effects of northbound long-haul international air travel on sleep quantity and subjective jet lag and wellness in professional Australian soccer players. Int J Sports Physiol Perform. 10(5):648–654.
Fowler P, Duffield R, Vaile J. 2014. Effects of domestic air travel on techincal and tactical performance and recovery in soccer. Int J Sport Physiol Perform. 9(3):378–386.
Fowler P, Duffield R, Waterson A, Vaile J. 2015b. Effects of regular away travel on training loads, recovery, and injury rates in professional Australian soccer players. Int J Sports Physiol Perform. 10(5):546–552.

Fowler P, Knez W, Crowcroft S, Mendham A, Miller J, Sargent C, … Duffield R. 2017. Greater effect of east versus west travel on jet lag, sleep, and team sport performance. Med Sci Sports Exerc. 49(12):2548–2561.

Fullagar H, McCunn R, Murray A. 2017. An updated review of the applied physiology of American Collegiate Football: the physical demands, strength/ conditioning,nutritional considerations and injury characteristics of America's favourite game. Int J Sports Physiol Perform. *[Epub ahead of print]*.

Hopkins W. 2004. How to interpret changes in an athletic performance test. Sportscience. 8(1):1–7.

Jehue R, Street D, Huizenga R. 1993. Effect of time zone and game time changes on team performance: National Football League. Med Sci Sports Exerc. 25(1):127–131.

Jones M. 2017. Differences in home advantage between sports. Psychol Sport Exerc. 34:61–69.

Lopez M, Matthews G, Baumer B (2017). How often does the best team win? A unified approach to understanding randomness in North American sport. https://arxiv.org/abs/1701.05976.

Nevill A, Holder R. 1999. Home advantage in sport: an overview of studies on the advantage of playing at home. Sports Med. 28(4):221–236.

Nevill A, Holder R, Bardsley A, Calvert H, Jones S. 1995. Identifying home advantage in international tennis and golf tournaments. J Sports Sci. 15:437–443.

Playoff CF. 2012. College football playoff selection committee protocol. http://www.collegefootballplayoff.com/documents/2017/10/20//CFP_Selection_Committee_Protocol.pdf?id=23.

Pollard R. 1986. Home advantage in soccer: a retrospective analysis. J Sports Sci. 4(3):237–248.

Pollard R, Gómez M. 2015. Comparison of home advantage in college and professional team sports in the United States. Coll Antropol. 39(3):583–589.

Pollard R, Pollard G. 2005. Long-term trends in home advantage in professional team sports in North America and England (1876-2003). J Sports Sci. 23(4):337–350.

Redding R. 2017. NCAA Football: 2016 and 2017 rules and intepretations. Indianapolis (IND): National Collegiate Athletic Association.

Robertson S, Joyce D. 2015. Informing in-season tactical periodisation in team sport: development of a match difficulty index for Super Rugby. J Sports Sci. 33(1):99–107.

Schwartz B, Barsky S. 1977. The home advantage. Soc Forces. 55:641–661.

Snyder E, Purdy D. 1985. The home advantage in collegiate basketball. Sociol Sport J. 2(p):352.

Tenenbaum G. 1996. Theoretical and practical considerations in investigating motivation and discomfort during prolonged exercise. J Sports Med Phys Fitness. 36(3):145–154.

Thirer J, Rampey M. 1979. Effects of abusive spectator behaviour on the performance of home and visiting intercollegiate basketball teams. Percept Mot Skills. 48:1047–1053.

Thomas S, Reeves C, Bell A. 2008. Home advantage in the Six Nations Rugby Union tournament. Percept Mot Skills. 106(1):113–116.

Wang W, Johnston R, Jones K. 2011. Home advantage in American College Football Games: a multilevel modelling approach. J Quantitative Anal Sports. 7(3):23.

Winter W, Hammond W, Green N, Zhang Z, Bliwise D. 2009. Measuring circadian advantage in Major League Baseball: a 10-year retrospective study. Int J Sports Physiol Perform. 4(3):394–401.

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